Universal Gravitation and Satellites

Answer key

Conventions used throughout

• **Universal gravitation:** $F_g = Gm_Am_B/r^2$, where r is the centre-to-centre distance and $G = 6.67 \times 10^{-11}$ N·m^2/kg^2.

• **Field strength:** $g = Gm_{planet}/r^2$, so $F_g = mg$ anywhere.

• **Earth:** $m_E = 5.98 \times 10^{24}$ kg, $r_E = 6.38 \times 10^6$ m (the standard textbook values).

• **Orbits:** gravity is the centripetal force, $F_c = F_G$. Every step keeps full precision; only final answers are rounded.

CONCEPTS 1–2

Black holes and buying gold

1. As a star ages, it is believed to undergo a variety of changes. One of the last phases of a star's life is to gravitationally collapse into a black hole. What will happen to the orbit of the planets of the solar system if our star (the Sun shrinks into a black hole)? (And of course, this assumes that the planets are unaffected by prior stages of the Sun's evolving stages.)

2. Having recently completed his first Physics course, Noah Formula has devised a new business plan based on his teacher's *Physics for Better Living* theme. Noah learned that objects weigh different amounts at different distances from Earth's center. His plan involves buying gold by the weight at one altitude and then selling it at another altitude at the same price per weight. Should Noah buy at a high altitude and sell at a low altitude or vice versa?

1. What will happen to the orbit of the planets if the Sun shrinks into a black hole?

Nothing changes. The force on each planet depends only on the two masses and the centre-to-centre distance. Shrinking the Sun changes neither: its mass stays the same and each planet is still the same distance from the Sun’s centre.

$$F_g = \frac{Gm_{Sun}m_{planet}}{r^2} \quad (m_{Sun},\ m_{planet} \text{ and } r \text{ all unchanged})$$

2. Should Noah buy at a high altitude and sell at a low altitude or vice versa?

Weight is $F_g = Gm_Em/r^2$, so the same gold weighs less far from Earth's centre. Buying by weight up high, Noah gets more gold for each newton he pays for. Selling down low, that same gold weighs more, so he is paid for more newtons.

FINAL ANSWER

1. The orbits stay the same (same mass, same distance)

2. Buy at a high altitude (gold weighs less) and sell at a low altitude (it weighs more)

CONCEPTS 3–4

Changing Earth's mass; weight on Jupiter

3. Anita Diet is very concerned about her weight but seldom does anything about it. After learning about Newton's law of universal gravitation in Physics class, she becomes all concerned about the possible affect of a change in Earth's mass upon her weight. During a (rare) free moment at the lunch table, she speaks up "How would my weight change if the mass of the Earth increased by 10%?" How would you answer Anita?

4. When comparing mass and size data for the planets Earth and Jupiter, it is observed that Jupiter is about 300 times more massive than Earth. One might quickly conclude that an object on the *surface* of Jupiter would weigh 300 times more than on the surface of the Earth. For instance, one might expect a person who weighs 500 N on Earth would weigh 150000 N on the *surface* of Jupiter. Yet this is not the case. In fact, a 500-N person on Earth weighs about 150 N on the *surface* of Jupiter. Explain how this can be.

3. How would my weight change if the mass of the Earth increased by 10%?

Weight is directly proportional to Earth's mass (r stays the same):

$$F_g = \frac{Gm_Em}{r^2} \quad \Rightarrow \quad F_{g,new} = \frac{G(1.10\,m_E)m}{r^2} = 1.10\,F_g$$

Her weight would also go up by 10% (her mass wouldn't change).

4. Explain how this can be.

Weight depends on the planet's radius as well as its mass, and it is divided by r^2. Jupiter's radius is about 11 times Earth's, so r^2 is about 125 times bigger. The two effects nearly cancel:

$$\frac{F_{g,Jupiter}}{F_{g,Earth}} = \frac{300}{(11)^2} \approx 2.5 \quad \Rightarrow \quad 500\ \text{N} \times 2.5 \approx 1.2 \times 10^3\ \text{N}$$

So the weight is far less than 150 000 N.

FINAL ANSWER

3. Her weight would increase by 10% ($F_g \propto m_E$)

4. Jupiter's much larger radius (about 11 times Earth's) divides the weight by about $11^2 = 125$, so the weight is only a few times bigger, not 300 times

Worksheet answer: 4. "a 500-N person on Earth weighs about 150 N on the surface of Jupiter"
Why the answers differ: The explanation (the larger radius) is right, but the number 150 N is wrong. Jupiter's surface gravity is about 24.8 N/kg, about 2.5 times Earth's 9.81 N/kg, so a 500 N person would weigh about 500 N × 2.5 ≈ 1.2 × 10^3 N there, **more** than on Earth. 150 N would mean Jupiter pulls more weakly than Earth, which is false. The estimate 300 ÷ 11^2 ≈ 2.5 above gives the same result.

SURFACE GRAVITY

Force of gravity on the surface of the Earth

For most of the problems where you have had to calculate the force of gravity, you have found the force between the earth and an object on (or very close to) the surface of the earth. In this case, fill in the values using Table 6 on page 710 of your text:

m_{earth} = _____ r = _____

Use these values to find a short-cut to calculating the force of gravity on an object of mass m_o when it is on the surface of the earth.

Put Earth's mass and radius into the law of universal gravitation. Everything except m_o is a constant, so work it out once:

$$m_{earth} = 5.98 \times 10^{24} \text{ kg} \qquad r = 6.38 \times 10^{6} \text{ m}$$

$$F_g = \frac{Gm_{earth}m_o}{r^2} = \left(\frac{Gm_{earth}}{r^2}\right)m_o$$

$$\frac{Gm_{earth}}{r^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.38 \times 10^{6})^2} = 9.799 \text{ N/kg}$$

So $F_g = (9.80 \text{ N/kg})m_o$, which is $F_g = mg$.

FINAL ANSWER

$m_{earth} = 5.98 \times 10^{24}$ kg, $r = 6.38 \times 10^{6}$ m

Short-cut: $F_g = (Gm_{earth}/r^2)m_o = (9.80 \text{ N/kg})m_o$

Satellites and space stations

a) For any satellite, what is the force that keeps it in its orbit?

b) To analyze the motion of a satellite, we combine Newton's Law of Universal gravitation with the equations for centripetal force. Sub in the **equations** for the two forces below: $F_c = F_G$

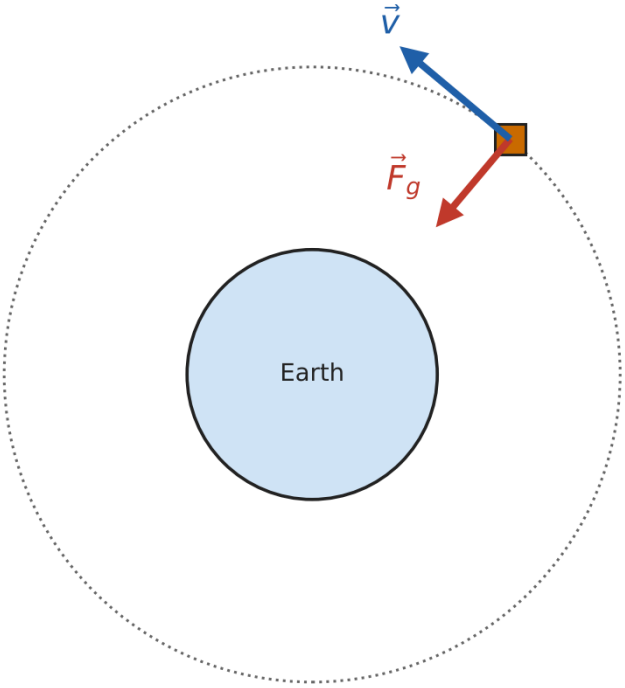
c) Starting with your equation above, derive an expression for the speed of the satellite orbiting the earth.

d) Derive an expression for the radius of a satellite's orbit around Earth in terms of the **period** of revolution using the appropriate expression for the centripetal acceleration.

a) For any satellite, what is the force that keeps it in its orbit?

Gravity from Earth, pulling the satellite toward Earth's centre. It is the only force acting.

A satellite in orbit



gravity is the only force: it points to Earth's centre

b) Sub in the equations for the two forces

$$F_c = F_G \quad \Rightarrow \quad \frac{mv^2}{r} = \frac{Gm_E m}{r^2}$$

c) Derive an expression for the speed of the satellite orbiting the earth.

Divide both sides by m: $\frac{v^2}{r} = \frac{Gm_E}{r^2}$

Multiply both sides by r: $v^2 = \frac{Gm_E}{r}$

Square root both sides: $v = \sqrt{\frac{Gm_E}{r}}$

d) Derive an expression for the radius of a satellite's orbit around Earth in terms of the period.

Use the period form of centripetal acceleration, $a_c = 4\pi^2 r/T^2$:

$$\frac{m4\pi^2 r}{T^2} = \frac{Gm_E m}{r^2}$$

Divide by m and multiply by r^2T^2: $4\pi^2 r^3 = Gm_E T^2$

Divide by $4\pi^2$ and cube root: $r = \sqrt[3]{\frac{Gm_E T^2}{4\pi^2}}$

FINAL ANSWER

a) Earth's gravity (toward Earth's centre)

b) $mv^2/r = Gm_E m/r^2$

c) $v = \sqrt{(Gm_E/r)}$

d) $r = \sqrt[3]{(Gm_E T^2 \div 4\pi^2)}$

EXAMPLE

Hubble Space Telescope

The Hubble Space Telescope is in a roughly circular orbit 598 km above the Earth's surface. Find (a) its speed in orbit; (b) its orbital period. HINT: you must consider the centre-to-centre distance between the Earth and the telescope.

Step 1 – Centre-to-centre distance

$$r = r_E + h = 6.38 \times 10^6 + 5.98 \times 10^5 = 6.978 \times 10^6 \text{ m}$$

(a) Speed

Use the result from Satellites (c):

$$v = \sqrt{\frac{Gm_E}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.978 \times 10^{6}}} = 7560 \text{ m/s}$$

(b) Period

One orbit is 2πr at speed v:

$$T = \frac{2\pi r}{v} = \frac{2\pi(6.978 \times 10^{6})}{7560} = 5799 \text{ s}$$

FINAL ANSWER

(a) v = 7.56 × 10^3 m/s

(b) T = 5.80 × 10^3 s (about 97 minutes)

Worksheet answer: a) 7559 m/s; b) 5793 s
Why the answers differ: Neither answer is wrong; they use different Earth data. The worksheet's numbers come from m_E = 5.97 × 10^{24} kg and r_E = 6.371 × 10^6 m (these give exactly 7559 m/s and 5793 s). This key uses the textbook values 5.98 × 10^{24} kg and 6.38 × 10^6 m (see Conventions), which give 7560 m/s and 5799 s. The two results differ by about 0.1%.

CONTINUOUS FREEFALL

Always falling, but missing the Earth

When the space station (or any satellite) orbits the earth, it travels at the appropriate speed that keeps it approximately the same distance from Earth's centre. As it circles Earth, it is undergoing **constant free fall** toward Earth.

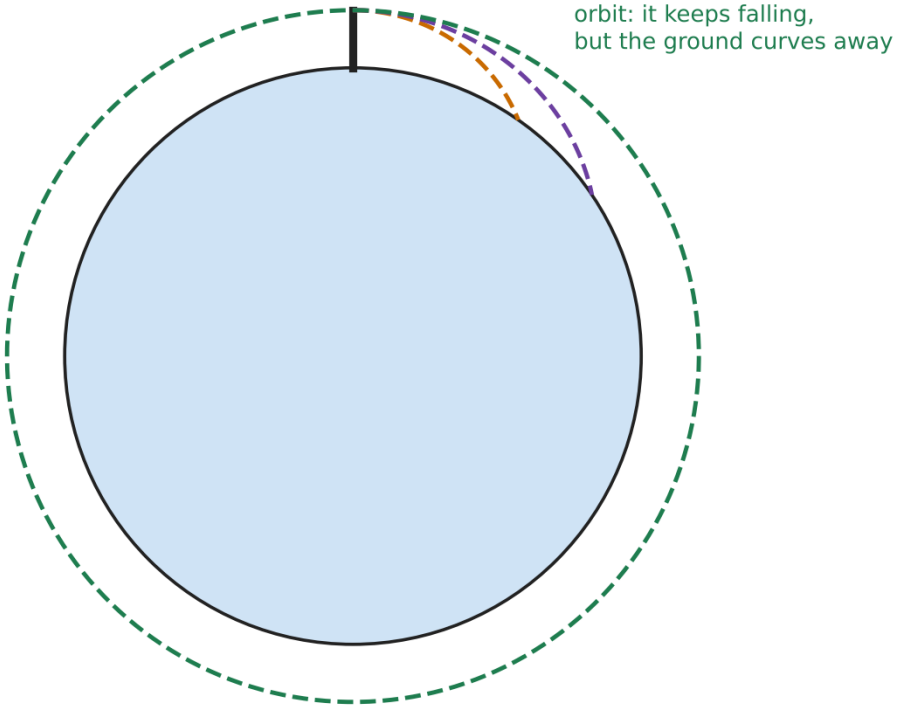
a) A satellite is "always falling, but missing the earth". Explain what this means in terms of orbits.

b) How do you think an astronaut on the ISS would feel based on this idea?

a) Explain what this means in terms of orbits.

Gravity is the only force on the satellite, so it is always falling toward Earth. But it also moves sideways very fast. While it falls, the curved surface of Earth drops away beneath it by the same amount, so it never gets closer. A slower object curves down and hits the ground; at orbital speed the fall exactly follows Earth's curve.

Faster and faster launches (Newton's cannon)



b) How do you think an astronaut on the ISS would feel based on this idea?

Weightless. The astronaut and the station fall together at the same rate, so the floor doesn't push on the astronaut ($F_N = 0$). Gravity still acts (about 90% as strong as on the ground), but with no normal force they feel no weight, just like in a freely falling elevator (2.6).

FINAL ANSWER

a) It falls toward Earth the whole time, but its sideways speed is so large that Earth's surface curves away as fast as it falls

b) Weightless: the astronaut and the station fall together, so there is no normal force